PMP® Exam Quick Reference

Key takeaways, frameworks, and formulas from the study deck and guide · aligned to the PMP® Examination Content Outline (July 2026)

The exam at a glance

Item	What to know
Format	180 questions (170 scored + 10 unscored pretest) in 240 minutes: a case-study section, then two parts of independent questions, with two 10-minute breaks. Review answers before each break; you can't return to a finished section.
Domains	**Process 41%** (10 tasks) · **People 33%** (8 tasks) · **Business Environment 26%** (8 tasks). Every exam covers all 26 tasks; enablers are examples, not a checklist.
Approach	About 40% predictive and 60% agile or hybrid questions, mixed through every domain.
Retakes	Up to three attempts in the one-year eligibility period.

Key takeaways: how PMI wants you to think

- **Assess before you act.** Gather facts, analyze impact, and consult the plan before choosing an action; the "first" step is usually to understand the situation.
- **Follow the process.** Changes to baselines go through integrated change control: analyze impact, get approval (CCB or delegated authority), then implement, update documents, and communicate. Never implement an unapproved change.
- **Lead as a servant leader.** Empower the team, coach, and remove impediments; the team decides how to do the work. Resolve problems with the team before escalating.
- **Escalate only beyond tolerance.** Within thresholds, work with the team; beyond them, escalate to the stakeholder with authority.
- **Deliver value early and often.** An agile team's highest priority is customer value: prioritize the backlog, release in increments (MVP), and maximize the work not done.
- **Engage stakeholders continuously.** Identify them early, analyze power and interest, tailor communication, and revisit the analysis as people and attitudes change.
- **Know risk from issue.** Might happen: risk register and a planned response. Has happened: issue log, an owner, and a workaround if no plan exists.
- **Prevent rather than inspect.** The most expensive defect is the one the customer finds; fix problems at the source (jidoka) and improve continuously (kaizen).
- **The structure tells you who to ask.** In functional and weak matrix organizations, negotiate with functional managers; in strong matrix and projectized ones, the PM has the authority.
- **Change happens one person at a time.** Build awareness and desire before training, and reinforce after go-live.
- **Close formally, every time.** Close contracts first, capture lessons learned throughout, archive records, and close even a cancelled project.

What each domain covers

Domain	Tasks (July 2026 outline)
People 33%	Develop a common vision · manage conflicts · lead the project team · engage stakeholders · align stakeholder expectations · manage stakeholder expectations · help ensure knowledge transfer · plan and manage communication
Process 41%	Integrated plan and delivery approach · scope · value-based delivery · resources · procurement · finance · quality · schedule · evaluate project status · project closure
Business Environment 26%	Governance · compliance · change control · impediments and issues · risk · continuous improvement · organizational change · external business environment changes

© 2026 jmtibbetts (LinuxNinja). All rights reserved. With credit to amarin@innodata.com.

Frameworks and models

Framework	Key idea	Exam cue
Tuckman	Forming, storming, norming, performing, adjourning	Storming is normal: coach, don't replace people; a team charter sets ground rules
Situational leadership	Directing, coaching, supporting, delegating, matched to competence and commitment	Adapt style to the person and task
Psychological safety	Inclusion, learner, contributor, challenger safety	Safety comes before candor and innovation
Conflict	Five levels: problem, disagreement, contest, crusade, world war; collaborate/problem solve is preferred	Diagnose the level from the language used, then de-escalate
Motivation	Maslow's needs; Herzberg hygiene vs. motivators; McGregor Theory X/Y	Pay and perks prevent dissatisfaction but don't motivate; agile assumes Theory Y
RACI and MBO	Responsible, Accountable, Consulted, Informed; goals cascade into agreed objectives	Exactly one A per activity
Stakeholder models	Six-step cycle; power/interest grid; salience (power, legitimacy, urgency); engagement matrix (unaware → leading, C vs. D)	High power, high interest: manage closely; the C–D gap drives the engagement plan
Knowledge	Explicit (written, like a recipe) vs. tacit (in people's heads)	Tacit knowledge transfers through people: mentoring, pairing
Schedule	Precedence (FS, SS, FF, SF); critical path = longest path, zero float; Gantt chart (team) vs. milestone chart (sponsors); schedule baseline	Crashing adds cost, fast tracking adds risk; change the baseline only through change control
Quality	Cost of quality: prevention and appraisal vs. internal and external failure; control charts; fishbone (6 Ms)	Rule of seven: seven points on one side of the mean signal a special cause
Procurement	Contract types: CR (buyer risk), T&M (shared), FP (seller risk); RFI, RFQ, RFP; bidder conferences	FFP is most common; bidder conferences exist for transparency
Estimating	Analogous (rough but quick), parametric, bottom-up (most accurate), three-point; ROM −25% to +75%, definitive −5% to +10%	Early with little detail: ROM; agile teams estimate relatively
Status and artifacts	Work performance data → information → reports; SMART measures; leading vs. lagging indicators; version control	Documents are not the plan; outdated documents call for version control
Closing	Confirm acceptance → close procurements → lessons learned → final report → transition → archive and release	A cancelled project still closes formally
Organization	OPAs (processes, corporate knowledge base) vs. EEFs (internal, external); functional, weak/balanced/strong matrix, projectized	Functional works in silos; projectized can cause job insecurity; matrix is a hybrid
Change control	Request → impact → CCB → update → implement → communicate; corrective, preventive, defect repair, update	Adaptive change goes to the backlog; the sprint is protected
Risk	Seven processes; threats: escalate, avoid, transfer, mitigate, accept; opportunities: escalate, exploit, share, enhance, accept; risk averse to risk seeking	Playing it safe is risk averse; learn appetites early to set thresholds
Impediments	Impediment (slows), obstacle (delays), blocker (stops); intervention strategy	Resolve standup impediments after the meeting
Organizational change	Lewin (unfreeze, change, refreeze); Kotter's 8 steps (organization); ADKAR (individual)	Trained but resisting: the barrier is Desire, not Knowledge
Environment	PESTLE: political, economic, social, technological, legal, environmental; strategy pyramid; ethics: responsibility, respect, fairness, honesty	When the external environment changes, assess the impact on scope or backlog first
Agile Manifesto	Individuals and interactions, working software, customer collaboration, responding to change (over the right-hand items); 12 principles	Simplicity: maximize the work not done
Scrum	Roles: product owner, scrum master, developers; 5 events; artifacts and commitments (product goal, sprint goal, definition of done)	PO decides what; developers decide how much and how
Kanban, lean, metrics	Limit WIP, manage flow; jidoka and kaizen; burndown (iteration), burnup (release), velocity, cumulative flow	A widening CFD band is a bottleneck; velocity forecasts, never compares teams

© 2026 jmtibbetts (LinuxNinja). All rights reserved. With credit to amarin@innodata.com.

Formulas

Group	Formula	Read it as
Earned value	**CV = EV − AC**	Positive: under budget
	SV = EV − PV	Positive: ahead of schedule
	CPI = EV / AC	Above 1: under budget
	SPI = EV / PV	Above 1: ahead of schedule
Forecasting	**EAC = BAC / CPI**	Typical: today's efficiency continues
	EAC = AC + (BAC − EV)	Atypical: the variance was a one-off
	EAC = AC + (BAC − EV) / (CPI × SPI)	Cost and schedule both drive the remaining work
	ETC = EAC − AC	Still to spend
	VAC = BAC − EAC	Negative: over budget at completion
	TCPI = (BAC − EV) / (BAC − AC)	Efficiency needed on the remaining work to hit BAC
Estimating	**PERT = (O + 4M + P) / 6**	Weighted (beta) average
	Triangular = (O + M + P) / 3	Simple average
	σ = (P − O) / 6	PERT standard deviation; variance = σ^2
Schedule	**TF = LS − ES = LF − EF**	Total float; zero on the critical path
	FF = ES(successor) − EF − 1	Free float with day-1 numbering (drop the − 1 from day 0)
	ES = EF(pred.) + 1; EF = ES + d − 1	Forward pass, day-1 numbering
Agile	**Iterations = points left / velocity**	Use a recent average velocity
	Cycle time = WIP / throughput	Little's Law: less WIP, faster delivery
Communication	**Channels = n(n − 1) / 2**	n includes the PM; 10 → 12 people adds 66 − 45 = 21

Worked example

BAC $100,000; PV $50,000; EV $40,000; AC $45,000. CV = −$5,000 (over budget); SV = −$10,000 (behind); CPI = 0.89; SPI = 0.80; EAC = $112,500; ETC = $67,500; VAC = −$12,500; TCPI = 60,000 / 55,000 = 1.09.

Memory aids

- Every variance and index starts with **EV**. Variances subtract; indices divide. Cost uses **AC**; schedule uses **PV**.
- Negative or below 1 is bad news; positive or above 1 is good news. CV and SV both positive: under budget and ahead of schedule.
- At project end SV returns to 0 (EV = PV = BAC), so use SPI or the schedule itself to judge a late finish.
- Estimate ranges: ROM −25% to +75%; budget −10% to +25%; definitive −5% to +10%.

Sources: PMI, PMP® Examination Content Outline (July 2026); PMI, PMBOK® Guide — Sixth and Seventh Editions; PMI and Agile Alliance, Agile Practice Guide (2017); The Scrum Guide (2020); framework authors as credited in the study deck; adapted from PMI course material.

© 2026 jmtibbetts (LinuxNinja). All rights reserved. With credit to amarin@innodata.com.